

Strategic coexistence theory for evolutionary game theory

Sang Woo Park^{1,*}

¹ School of Biological Sciences, Seoul National University, Seoul, Korea

*Corresponding author: sangwoopark@snu.ac.kr

Abstract

Introduction

Understanding the mechanisms that promote evolutionary stability of a behavioral strategy is a fundamental aim in evolutionary biology. To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition via payoff inequalities, invasion criteria, and stability analysis. However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Tarnita and Traulsen, 2025), even though the classic replicator equation stems from the generalized Lotka-Volterra equation.

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology: the dominance of a singular strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a singular strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition? Even though EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanism between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms. MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast,

fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains. Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover

Strategic coexistence theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j . However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

71 When $1 - \rho > 0$, each strategy performs better, on average, in the environment
 72 generated by its competitor than in the environment generated by itself. In other
 73 words, intraspecific competition is stronger than interspecific competition.

74 Second, we define the fitness difference as the ratio between the geometric means
 75 of the multiplicative payoff of each strategy across the two single-strategy environ-
 76 ments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

77 In other words, strategy j has an average competitive advantage over strategy i
 78 ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated
 79 by itself and its competitor. Then, analogous to MCT, mutual invasion requires the
 80 fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

81 This decomposition allows us to understand how different mechanisms contribute to
 82 the coexistence or exclusion of competing strategies.

83 Two strategy games

84 As an illustrative example, we begin with classic two-strategy games, which can be
 85 described by the following payoff matrix (Figure 1A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

86 Here, the first and second rows represent strategies 1 and 2, respectively, and the
 87 columns represent the strategy of the interacting opponent. Writing x_1 to denote
 88 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

89 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

90 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
 91 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
 92 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
 93 the game space into four regimes (Figure 1B): prisoner's dilemma ($T > R$, $P >$
 94 S), hawk-dove game ($T > R$, $S > P$), stag-hunt game ($R > T$, $P > S$), and

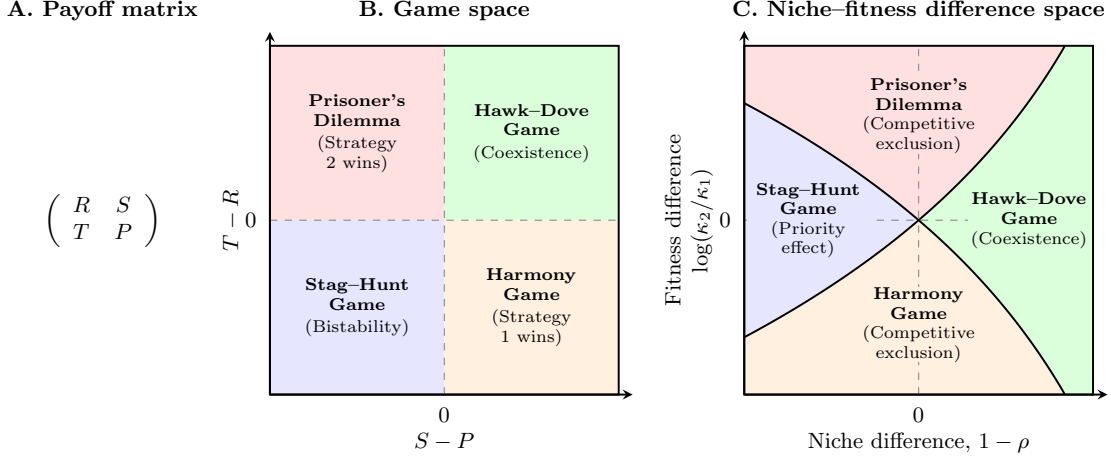


Figure 1: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime.

harmony game ($R > T$, $S > P$). In standard EGT, these outcomes are obtained by analyzing the stability of equilibrium conditions. For simple games, this analysis is straightforward.

A coexistence-theoretic approach recovers the same classification but provides a different interpretation (Figure 1C). For a two-strategy game determined by this payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy can invade when rare and the two strategies coexist. The remaining games also have natural interpretations in niche-fitness difference space. Prisoner's dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag-hunt games

107 represent a priority-effect regime, where neither strategy can invade the other and
 108 the outcome depends on initial conditions. Therefore, coexistence theory not only
 109 allows us to recover the classic results from two-strategy games but also provides new
 110 interpretations for these results in terms of niche and fitness differences. Building on
 111 this, we next apply this framework to microbial cooperation, where nonlinear benefits
 112 and environmental context can alter the coexistence of cooperators and defectors.

113 Microbial cooperation and public goods

114 Here, we consider a simple, phenomenological model of competition between cooper-
 115 ating and cheating yeast in a cooperative sucrose-metabolism system, where invertase
 116 production by cooperators allows cheaters to utilize glucose, a public good released
 117 through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by
 118 cooperators incurs a cost c and generates a total benefit of 1 that is captured pri-
 119 vately with efficiency ϵ . Assuming a linear relationship between glucose availability
 120 and microbial growth, the payoffs for cooperators π_C and cheaters (defectors) π_D can
 121 then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

122 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 123 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

124 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 125 f (Figure 2A). Therefore, cooperation is favored when efficiency is greater than the
 126 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 127 efficiency, $c > \epsilon$ (Gore et al., 2009).

128 Our framework allows us to reinterpret this result from the perspective of coexis-
 129 tence theory. In particular, the two strategies have no niche difference, corresponding
 130 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 131 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

132 Therefore, this model always predicts competitive exclusion, except in the neutral
 133 case $c = \epsilon$. The outcome is determined entirely by the fitness difference between
 134 cooperators and defectors, rather than by stabilizing niche differences.

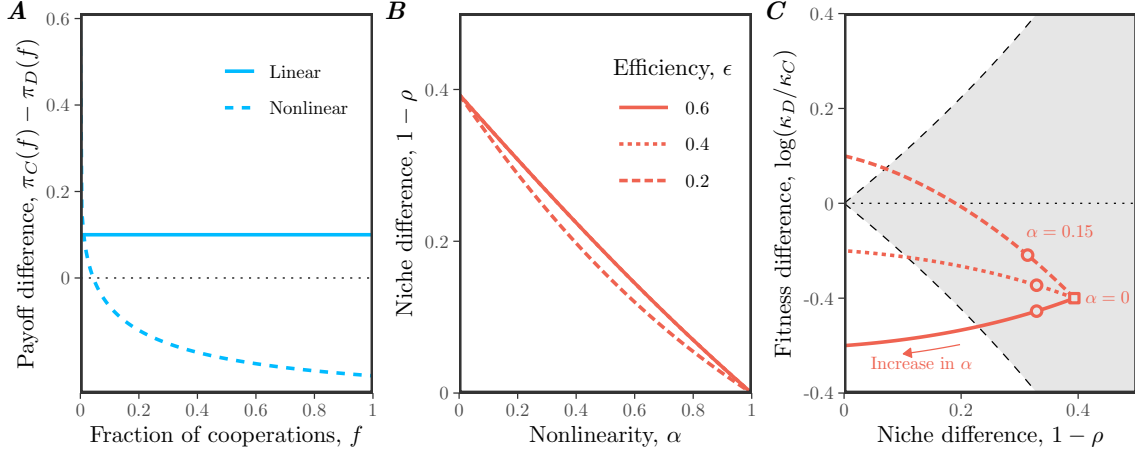


Figure 2: Caption. (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.2$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha = 0$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.2$, while allowing ϵ and α to vary.

In real biological systems, the effect of glucose on microbial growth can follow a nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - f(1 - \epsilon)^\alpha]. \quad (19)$$

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure 2A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, we can show that the niche and fitness differences for this model are given

147 by

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

148 These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For
149 example, the cost of cooperation does not affect niche difference. Instead, niche
150 difference is generated by the concave relationship between glucose availability and
151 microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0
152 (Figure 2B,C). In contrast, as $\alpha \rightarrow 0$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

153 which no longer depends on the private capture efficiency ϵ (Figure 2C). Thus, strong
154 concavity not only generates stabilizing niche differences, but also acts as an equal-
155 izing mechanism (Figure 2C).

Supplementary Text

S1 Derivation of the strategic coexistence theory

To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

where x_i represents the frequency of strategy i , π_i represents the payoff of strategy i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

we obtain the following inequality that defines conditions for mutual invasion and long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

169 S2 Applications to classic two strategy games

170 Consider a two strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

171 There are four classic games that we can define from this simple payoff matrix:
 172 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 173 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 174 each game corresponds to a distinct region in the niche-fitness difference space under
 175 SCT.

176 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 177 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 178 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

179 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

180 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

181 As before, SCT allows us to predict the mutual invasion of two strategy using the
 182 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

183 Then, simple algebra reveals that each condition in mutual invasion inequality
 184 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

185 **References**

- 186 Gore, J., H. Youk, and A. Van Oudenaarden (2009). Snowdrift game dynamics and
187 facultative cheating in yeast. *Nature* 459(7244), 253–256.
- 188 Tarnita, C. E. and A. Traulsen (2025). Reconciling ecology and evolutionary game
189 theory or “When not to think cooperation”. *Proceedings of the National Academy*
190 *of Sciences* 122(14), e2413847122.